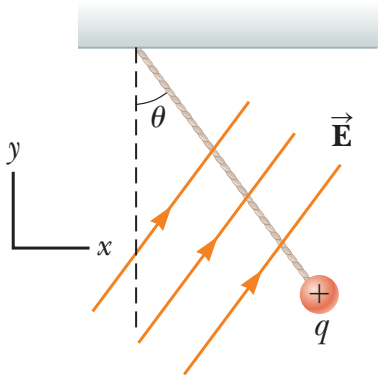
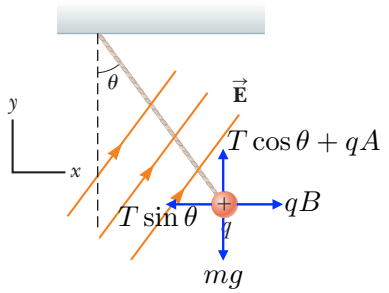
Problem 1 (30 points)



A charged cork ball of mass m is suspended on a light string in the presence of a uniform electric field as shown in the figure. When $\vec{E} = A\hat{\imath} + B\hat{\jmath}$, where A and B are positive quantities, the ball is in equilibrium at the angle θ.

(a) (15pts.) Find the charge on the ball in terms of m, A, B, θ and g.



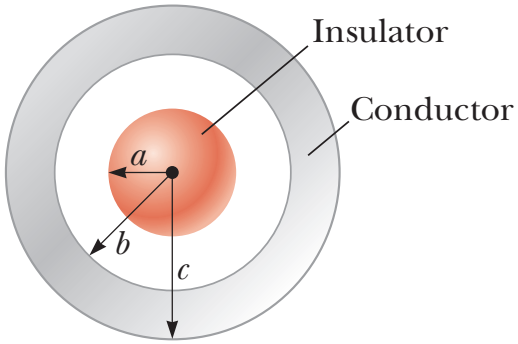
$T\cos\theta + qB = mg$
$T\sin\theta = qA$
$T = (qA)/(\sin\theta)$
$((qA)/(\sin\theta))(\cos\theta) + qB = mg$
$q = (mg)/(A\cot\theta + B)$

(b) (15pts.) Find the tension in the string in terms of m, A, B, θ and g.

From equations above it is easy to leave T alone.

$$T = \frac{mgA}{A\cos\theta + B\sin\theta}$$

Problem (2) (35 points)



A solid, insulating sphere of radius a has a uniform charge density throughout its volume and a total charge Q.

Concentric with this sphere is an uncharged, conducting, hollow sphere whose inner and outer radii are b and c as shown in the figure.

Find the electric field for the following regions:

(a) (10 pts.) $r < a$

For a Gaussian sphere with $r < a$ the charge enclosed is:
$Q_{enc} = \rho \frac{4}{3}\pi r^3$
where $\rho = \dfrac{Q}{\frac{4}{3}\pi a^3}$. By using Gauss' Law: $E(r)4\pi r^2 = \dfrac{Q_{\text{enc}}}{\epsilon_0} = \dfrac{Qr^3}{\epsilon_0 a^3}$

So, $\boxed{\vec{E}(r) = \dfrac{Q}{4\pi\epsilon_0}\dfrac{r}{a^3}\hat{r}}$

(b) (10 pts.) $a < r < b$

For a Gaussian sphere with $a < r < b$ the charge enclosed is:
$Q_{enc} = Q$
By using Gauss' Law: $E(r)4\pi r^2 = \dfrac{Q}{\epsilon_0}$

So, $\boxed{\vec{E}(r) = \dfrac{Q}{4\pi\epsilon_0}\dfrac{1}{r^2}\hat{r}}$

(c) (10 pts.) $b < r < c$

Electric field is 0 inside the conductor in electrostatic conditions. $\boxed{\vec{E}(r) = 0}$

(d) (5 pts.) Find the total charge on the inner and the outer surface of the conductor spherical shell.

If the electric field to be 0 inside the conductor, the inner surface must have a total charge of $-Q$. Therefore, the outer surface of the conductor must have $+Q$ since its total charge is zero.

Problem (3) (35 points)

For the same setup in Problem 2, find the electric potential as a function of r (the radial distance from the center of the spheres) for the following regions:
(Take $V(r \to \infty)$=0.)

(a) (5 pts.) $r > c$

$$V_r - V_\infty = \int_r^\infty \vec{E} \cdot \vec{r} = \frac{Q}{4\pi\epsilon_0} \int_r^\infty \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{r}\right)_r^\infty = \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{\infty} + \frac{1}{r}\right) \Longrightarrow \boxed{V(r) = \frac{Q}{4\pi\epsilon_0 r}}$$

(b) (10 pts.) $b < r < c$

This region is inside the conductor. Conductor itself is an equipotential surface that means everywhere on the conductor is at the same potential. And this potential can be found by using previous result with $r = c$. Therefore, $\boxed{V(r) = \frac{Q}{4\pi\epsilon_0 c}}$

(c) (10 pts.) $a < r < b$

$$V_r - V_\infty = (V_r - V_b) + (V_b - V_c) + (V_c - V_\infty)$$

$$V_r - V_\infty = \left(\int_r^b \frac{Q\ dr}{4\pi\epsilon_0 r^2}\right) + 0 + \frac{Q}{4\pi\epsilon_0 c} = \frac{Q}{4\pi\epsilon_0}\left(-\frac{1}{b} + \frac{1}{r}\right) + \frac{Q}{4\pi\epsilon_0 c} \Longrightarrow \boxed{V(r) = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{r} - \frac{1}{b} + \frac{1}{c}\right)}$$

(d) (10 pts.) $r < a$

From previous problem, we found that the electric field inside the insulating sphere is $\vec{E}(r) = \frac{Q}{4\pi\epsilon_0}\frac{r}{a^3}\hat{r}$.

$$V_r - V_\infty = (V_r - V_a) + (V_a - V_\infty)$$

where $(V_a - V_\infty)$ is the potential difference from part(c) with $r = a$.

$$V_r - V_\infty = \left(\int_r^a \frac{Q\ r\ dr}{4\pi\epsilon_0 a^3}\right) + \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b} + \frac{1}{c}\right) = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{2a} - \frac{r^2}{2a^3}\right) + \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b} + \frac{1}{c}\right)$$

$$\boxed{V(r) = \frac{Q}{4\pi\epsilon_0}\left(-\frac{r^2}{2a^3} + \frac{3}{2a} - \frac{1}{b} + \frac{1}{c}\right)}$$